RunCPM 7xtgv-terminal Tooling Guide

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To try the project, open the published browser application; no host tooling or native mirror is needed. For browser controls, start with the User Guide.

The host tools build/serve the browser application, build or upload MCU firmware, and connect a native terminal window to hardware or the SDL desktop emulator. This guide maps those responsibilities and the files which connect them. Use the Build Guide for setup and executable launch recipes, and the Reference Guide for keyboard and transfer syntax.

Native desktop emulation is buildable on Linux and Windows. Both use the same workflow: explicit setup, `desktop-rebuild`, then `desktop-local` or `desktop-card`. Linux launchers are under `tools/targets/` with `.sh` extensions; Windows launchers are under `tools-win/targets/` with `.cmd` extensions. The terminal/CP/M behavior is shared; host mirror and process transport differ. Hardware build/upload remains a Linux workflow. For first installation, use the Build Guide's setup steps.

Paths below are relative to `src/RunCPM_7xtgv-terminal/`.

1. Structure at a glance

Route	Execution and tools
Published browser	Open URL → page input + Wasm runtime/canvas → private memory card → downloaded ZIP. No Python mirror or hardware uploader.
Local browser development	`tools-web/setup.sh` → `desktop-rebuild.sh` → `desktop-local.sh`. One shared build; target selection in the page. Hosting/deployment details live in the tools-web README.
Native desktop	Host setup → target-specific `desktop-rebuild` → `desktop-local` or `desktop-card`. Native mirror input, SDL output and direct host-card writes.
Physical hardware	Linux target-specific `rebuild.sh` → `reflash.sh` → native mirror/serial connection. The MCU accesses its physical card.

Both hosts use the same desktop operations: rebuild, launch with the repository card, or launch with a selected external card. These diagrams show how each host connects those operations to the emulator and its mirror.

Linux host tooling

7xtgv-terminal · Linux host tooling

Launchers, shared mirror helpers, transport ownership and maintenance

Shared setup and target selection
setup.sh · terminal.conf · target_config.py · targets/<target>.json
Dependencies, target metadata, serial selection and host launch defaults

Hardware: selected board folder
`mirror.sh`

Desktop: selected board folder
`desktop-local.sh / desktop-card.sh`

target_mirror.py
Resolve serial endpoint; select visible mirror.
No firmware build or upload.

target_desktop.py → desktop.py
Start xtgv_desktop; obtain PTY; select mirror.
Uses target_mirror.mirror_command().

Shared mirror selection: one of three launchers

Konsole: kon.sh
7xtgv-terminal.profile
7xtgv-terminal.keytab

Kitty: kit.sh
kitty.conf
Keyboard-event mappings

Current terminal
cpm-session.sh
Uses the caller's terminal

Konsole profile/keymap identifier: 7xtgv-terminal. All three paths use the session helper.

cpm-session.sh · create or attach to tmux
Hardware session: 7xtgv-terminal
Desktop override: XTGV_SESSION=7xtgv-desktop-<PID>

terminal.py · PC screen, keyboard bridge and transfer panel
One connection to the selected CP/M endpoint.

USB serial ↔ MCU
RunCPM + terminal + TFT; MCU card storage.

PTY ↔ xtgv_desktop
RunCPM + terminal + SDL; selected host card.

Separate maintenance entry points

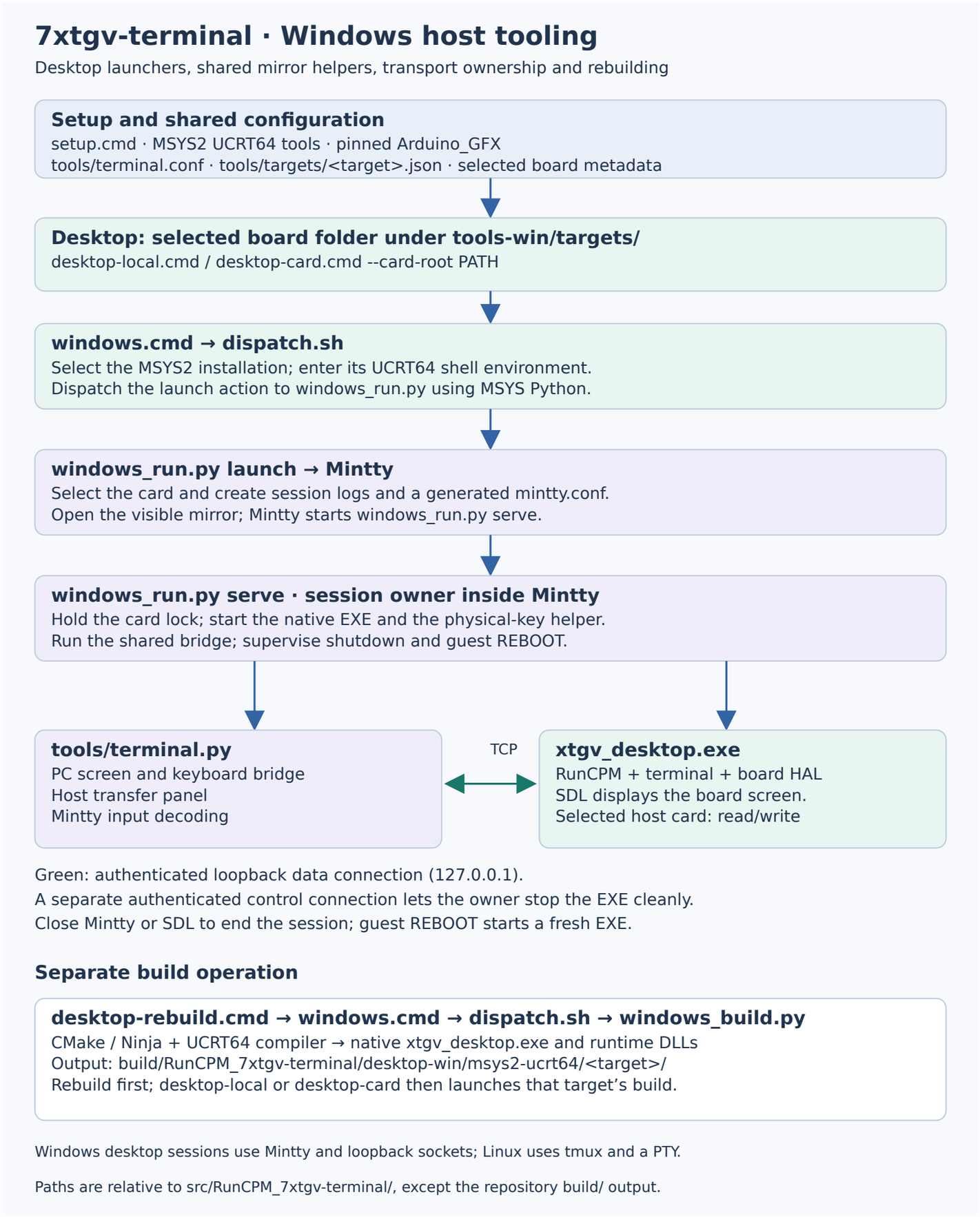
MCU build / upload
rebuild.sh → build/stage
reflash.sh → stop mirror,
upload, restore mirror

Desktop build
desktop-rebuild.sh
→ desktop.py build
→ CMake / host executable

Guest transfers
VTPUT/VTGET: existing bridge
Direct transfer: own connection
INI fetch: stop / fetch / restore

Paths are relative to src/RunCPM_7xtgv-terminal/. Blue = launch; green = byte transport.

Windows host tooling



Blue arrows show launch flow. Green arrows show the bidirectional connection between the bridge and CP/M. Maintenance/build boxes summarize separate operations; they are not additional stages of an ordinary mirror launch. Both hosts use `terminal.py` and the shared terminal protocol. Transport and process details are described in TRG: desktop host tooling.

2. Names and their roles

Name	Role
`tools/cpm-session.sh`	Shell helper which creates or attaches to a tmux session, starts the bridge and handles attachment cleanup. This is a script filename.
`7xtgv-terminal`	Default hardware tmux session name. The same spelling identifies the Konsole profile and keyboard table, but those are separate objects.
`tools/7xtgv-terminal.profile`	Konsole appearance/interaction settings. Its visible name is `7xtgv-terminal`; `KeyBindings=7xtgv-terminal` selects the keyboard table.
`tools/7xtgv-terminal.keytab`	Konsole keyboard table. It converts host chords into terminal key events and control sequences; application-specific interpretation is configured by the guest INI.
`tools/kon.sh`	Installs the profile and a configuration-specific generated keytab into the user's Konsole data directory, opens Konsole and invokes `cpm-session.sh`.
`tools/kit.sh`	Generates a configuration-specific copy of `tools/kitty.conf` in the host cache, opens Kitty with it and invokes the same `cpm-session.sh`.
`7xtgv-desktop-<PID>`	Isolated desktop tmux session name. The desktop launcher supplies it through `XTGV_SESSION`, replacing the hardware default for that process.
`tools/terminal.py`	PC mirror/bridge. Exchanges bytes with the MCU serial endpoint, Linux desktop PTY or Windows loopback endpoint and handles mirror controls and the host transfer panel.

Konsole installs the two files under `${XDG_DATA_HOME:-$HOME/.local/share}/konsole/` on each `kon.sh` launch. Selecting a Konsole profile does not select an application bank. The firmware or desktop terminal resolves bank settings from the selected card's `A0:7XTGVTRM.INI`.

`tools/generate_key_events.py` maintains the generated key-event blocks in `7xtgv-terminal.keytab`, `kitty.conf` and `kitty-cpm.conf`. Those host mappings and the terminal decoder must agree on event encoding. Normal `kon.sh`/`kit.sh` launches generate separate copies using the current host configuration; they leave the checked-in templates and existing windows unchanged. Konsole selects `7xtgv-terminal-<content-hash>.keytab` through a launch override. Kitty uses a content-specific file under `${XDG_CACHE_HOME:-~/.cache}/7xtgv-terminal/`.

3. Hardware mirror and ownership

1. The selected board folder's `mirror.sh` invokes `tools/target_mirror.py` with the board's canonical target name.
2. `target_config.py` loads `tools/targets/<target>.json` and resolves the serial endpoint. Launcher defaults and overrides also use `tools/terminal.conf`.
3. `target_mirror.py` chooses Konsole, Kitty or the current terminal and invokes `kon.sh`, `kit.sh` or `cpm-session.sh` respectively.
4. `cpm-session.sh` attaches to the existing shared `7xtgv-terminal` session, or creates it and starts `terminal.py`. The bridge is the serial reader; extra windows attach to tmux instead of opening another serial connection.

The creating mirror attachment owns session cleanup. Its exit terminates the host bridge; the MCU continues running CP/M. `cpm-session.sh --monitor` creates an observing attachment without that owning cleanup path.

The shell helper accepts `XTGV_SESSION` as an override. The hardware reflash and INI-fetch helpers operate on the standard `7xtgv-terminal` session, so custom hardware session names require corresponding lifecycle coordination. Desktop launchers supply their own isolated session explicitly.

4. Desktop launch and ownership

Use the same three operations in the chosen target folder on either host: `desktop-rebuild` builds, `desktop-local` opens the repository's `cpmcard/`, and `desktop-card --card-root PATH` opens a selected external card. Add `.sh` on Linux or `.cmd` on Windows. Setup and first-run commands are in the Build Guide.

Linux desktop mirror

a. Run `desktop-local.sh` or `desktop-card.sh --card-root PATH` from the target folder under `tools/targets/`.

b. The launcher starts the selected emulator and opens Konsole, Kitty or the current terminal as its mirror.

c. Type CP/M commands into the mirror; the SDL window shows the board display.

d. Close the owning mirror or SDL window to end the session. Guest `REBOOT` restarts the emulator.

Windows desktop mirror

a. Run `desktop-local.cmd` or `desktop-card.cmd --card-root PATH` from the target folder under `tools-win/targets/`.

b. The launcher starts the selected emulator and opens Mintty as its mirror.

c. Type CP/M commands into the mirror; the SDL window shows the board display.

d. Close the owning mirror or SDL window to end the session. Guest `REBOOT` restarts the emulator.

Shared session controls

Both launch modes open their selected card read/write, including guest files, INI and history. A per-card lock prevents another participating desktop from opening the same card for writing. Close the session before removing the card.

Both hosts accept `--window-scale N` (integer 1..9, default 1). For example, `desktop-local.sh --window-scale 3` or `desktop-local.cmd --window-scale 3` shows the entire fixed-size SDL panel at triple size. This is independent of terminal glyph zoom and does not change guest geometry or the INI.

Alt Q is application exit in the supplied Konsole, Kitty and Mintty profiles. Bank, brightness and zoom controls use the shared terminal mappings in the Reference Guide. `XTGV_HACK_TOPROW` in `tools/terminal.conf` controls optional number-row conversion; open a new mirror after changing it. Keyboard implementation and configuration precedence are described in TRG: desktop host tooling.

Choosing the Linux mirror

The default `--mirror auto` checks for a graphical session (`DISPLAY` or `WAYLAND_DISPLAY`), then chooses Konsole if it is on PATH, otherwise Kitty. If neither is available, or there is no graphical session, it uses the current terminal. With both installed, Konsole wins; the choice does not depend on which terminal launched the script.

Choose explicitly for a session:

```
./desktop-local.sh --mirror konsole
# Or choose Kitty instead:
./desktop-local.sh --mirror kitty
```

`desktop-card.sh` and the hardware `mirror.sh` accept the same option. `--mirror terminal` uses the current terminal. An explicitly requested missing terminal produces an error; it does not silently select another. Konsole has been the primary interactive Linux test path. Kitty has a supplied integration and configuration checks, with less physical-key/user testing; availability does not imply equal acceptance coverage. Windows uses Mintty.

See TRG for process ownership, transports and platform-specific launcher options, and the Windows tooling notes for logs and diagnostic commands.

Browser build and local hosting

`tools-web/desktop-rebuild.sh` builds one shared Wasm runtime for all six current displays. The common `tools-web/desktop-local.sh` launcher opens the standalone page; select the target there, without a per-board build or launcher. An optional `--target DISPLAY_ID` sets the initial selection.

The page handles keyboard input, integer canvas scale, text/image snapshots and card ZIP import/export. Switching targets restarts RunCPM while keeping the memory card; Stop and download its ZIP to keep changes beyond the page lifetime. The retained `--native-mirror` proof is optional and still starts ES3_35. Setup, checks and hosting details are maintained in the tools-web README.

Retired WezTerm experiment

The optional WezTerm launchers were removed on 261012 after Windows colour tests. Use Konsole/Kitty on Linux and Mintty on Windows; the shared host keyboard configuration remains supported. Historical findings are retained in the retirement record.

5. Builds, uploads and transfers

Entry point	Path and effect
`rebuild.sh`	`target_reflash.py --build-only` delegates to `reflash.py --build-only`. Builds/stages the selected MCU target under `build/`; does not stop the mirror or upload.
`reflash.sh`	`target_reflash.py --upload-only` selects staged firmware, stops the standard hardware mirror, uploads through `reflash.py` for serial targets or `bootsel_upload.py` for RP2350, then restores the selected visible mirror. Upload restarts the MCU and loses guest application state.
`desktop-rebuild.cmd`	`windows.cmd` / `dispatch.sh` delegates to `windows_build.py` and CMake in MSYS2 UCRT64. Produces the native Windows EXE and adjacent runtime DLLs; launch is separate.
`desktop-rebuild.sh`	`target_desktop.py build` delegates to `desktop.py build` and CMake. Produces a host executable; launch remains a separate operation.
Guest `VTPUT` / `VTGET`	The running mirror opens its transfer panel; `terminal.py` loads `tools/cpm-transfer/transfer.py` and uses the existing connection. No second serial reader is opened.
Direct `tools/cpm-transfer/transfer.py`	Owns its direct connection. Stop the hardware mirror first and restore its visible attachment afterward.

Entry point	Path and effect
`tools/fetch-7xtgvtrm-ini.sh`	Stops the standard hardware mirror, downloads device `A0:7XTGVTRM.INI`, archives the preceding local copy, then restores the Konsole mirror. This fetch does not upload the INI.
`tools-win/setup.cmd`	Explicit Windows prerequisite setup in the selected MSYS2 installation, plus project-local GFX sources. Run the full MSYS2 update separately first.
`tools/setup.sh`	Explicit Linux prerequisite installation and host setup. Rebuild/launch tools report missing prerequisites rather than installing them implicitly.

After a tooling-name update, close a hardware mirror started by the previous version before launching its replacement. A different tmux session name does not free a serial endpoint already owned by an older bridge. Existing Konsole windows retain their loaded profile; new `kon.sh` launches install/select the current profile. Old installed profiles can remain until no longer needed.

For terminal/core/HAL source ownership, see the Technical Reference Guide.